

L09: Advanced Deep Learning for Computer Vision

CSci 560 Deep Learning w/ Python (Chollet) Ch. 9

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Summer 2025

L09.1 Three Essential Computer Vision Tasks

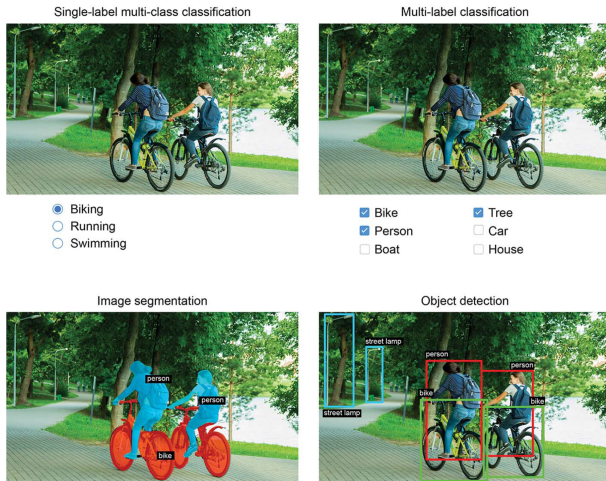
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The Three Main Computer Vision Tasks



- **Image classification** goal is to assign one or more labels to whole image.
- **Image segmentation** goal is to “segment” or “partition” an image into different areas.
- **Object detection** goal is to draw **bounding boxes** around objects of interest and classify.

Figure 1: The three main computer vision tasks: classification, segmentation, detection. (Chollet, 2021, pg.239)

L09.2 An Image Segmentation Example

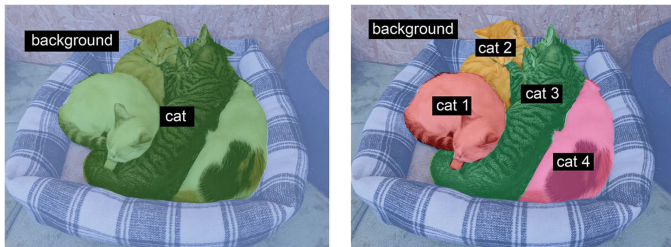
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Semantic vs. Instance Segmentation



- **Semantic segmentation** each pixel is independently classified.
- **Instance segmentation** seeks not only to classify image pixels by category, but also to parse out individual object instances.

Figure 2: Semantic segmentation vs. instance segmentation.
(Chollet, [2021](#), pg.241)

L09.3 Modern Convnet Architecture Patterns

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L09.4 Interpreting what Convnets Learn

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Chollet, F. (2021). *Deep learning with python* (second). Manning.